👨‍🔬 **Hands-On Lab for AWS Shield (Standard & Advanced)**

🛡️ AWS Shield – Hands-On Lab

🎯 Lab Objective

- Understand **AWS Shield Standard**
- Enable and explore **AWS Shield Advanced**
- Integrate with **CloudFront** and **WAF**
- Analyze **attack protection metrics/logs**
- (Optional) Simulate allowed traffic and observe metrics

🧪 Lab 1: Explore AWS Shield Standard (Free Protection)

🧰 Prerequisites:

- AWS Account with basic CloudFront and ALB setup

🧾 Steps:

1. **Launch a Static Website with CloudFront**
    - Create an S3 bucket (`shield-lab-site`)
    - Upload `index.html`

- Create a CloudFront distribution using the bucket as the origin

2. **Access AWS Shield**
   - Go to **AWS Console > Shield**
   - Confirm that **Shield Standard is automatically enabled** for CloudFront
3. **Open CloudWatch Metrics**
   - Go to **CloudWatch > Metrics > AWS/WAFV2 or Shield**
   - Observe default metrics like:
     - AllowedRequests
     - BlockedRequests
     - RequestCount
     - AttackVectors
4. **Perform Light Test**
   - Open multiple tabs of your CloudFront URL

Use curl with delays:

```
curl https://<your-cloudfront-domain> --limit-rate 100k --retry 5
```

5. **Expected Results:**
   - Shield silently protects from basic attacks (you won't be blocked).
   - Metric counts increase in CloudWatch.

🧪 Lab 2: Enable AWS Shield Advanced (Trial for 30 Days)

🧾 Steps:

1. **Go to AWS Shield Console**
   - Navigate to AWS Shield > Overview
   - Click **"Subscribe to Shield Advanced"**
   - Confirm the free trial (⚠️ billing applies after trial)
2. **Protect a Resource**
   - Choose:
     - **CloudFront distribution**
     - OR **Application Load Balancer**
     - OR **Elastic IP**
   - Click **"Add protection"**
   - Optionally enable **WAF association**
3. **Configure Notifications**
   - Create an **SNS topic** (e.g., shield-alerts)
   - Subscribe with your email
   - Shield will send alerts during attack detection

4. **Enable Logging (Optional)**
    - Choose an S3 bucket or CloudWatch Logs group
    - Logs include source IPs, attack type, duration

🧪 Lab 3: Integration with AWS WAF

1. **Attach WAF to CloudFront (or ALB)**
    - Use AWS WAF to define rules (e.g., block bad IPs or SQL injection)
    - Combine WAF for Layer 7 + Shield for L3/L4 defense

Simulate Application-level Request

curl "https://<cloudfront-url>?user=admin'--"

2.
3. **Monitor Logs**
    - Check WAF logs in S3 or CloudWatch
    - Confirm Shield did not block, but WAF did

🔬 Observation Tasks for Students

Task	Tool
View protection status	AWS Shield console

View real-time metrics	CloudWatch
Check DDoS threat trends	Shield Global Threat Dashboard
Review WAF logs for traffic	CloudWatch or S3
Create alerts	SNS + CloudWatch Alarms

📸 Deliverables / Reports (For Students)

Deliverable	What to Include
📷 Screenshot 1	Shield console showing Standard or Advanced active
📷 Screenshot 2	Protected resources with CloudFront or ALB
📷 Screenshot 3	CloudWatch metrics (request counts, attacks)
📄 Report	100–200 word reflection: What does Shield protect? How is it different from WAF?
📷 Optional	WAF logs showing simulated malicious traffic blocked

---

🧠 Lab Summary

- ✅ AWS Shield Standard: Always-on DDoS defense (free)
- ✅ AWS Shield Advanced: Enhanced protection + DRT + refund
- ✅ Works best with CloudFront, ALB, Route 53
- ✅ Integrates with AWS WAF and CloudWatch